MSc Finance 20127

Corporate Finance (Business Valuation)
02 — From Financial Statements to Free Cash Flows

Valuation Methods, Enterprise vs Equity Value, FCFO and FCFE

Stefano Rossi

Bocconi, IGIER, CEPR, and ECGI





Where We Are in the Valuation Workflow

Equity story $\longrightarrow$ **Financial statements** $\longrightarrow$ Forecasts $\longrightarrow$ **Cash flows** $\longrightarrow$ Discount rate $\longrightarrow$ Terminal value $\longrightarrow$ Multiples $\longrightarrow$ Value bridge $\longrightarrow$ Recommendation

Today: **Financial statements** $\longrightarrow$ **Cash flows** $\longrightarrow$ **Enterprise/Equity Value**

Today's Professional Task

Professional task:

- You receive a company's income statement, balance sheet, and cash-flow statement
- Your job is to produce valuation-ready FCFO and FCFE
  - separate operating, financing, and non-core items
  - compute cash flows consistently
  - avoid mixing enterprise value and equity value

Plan for Today

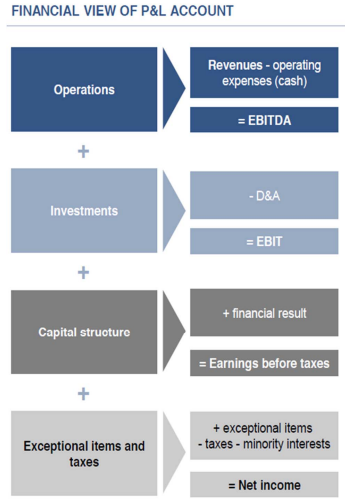
Layer	What We Learn	Why It Matters
CORE	Enterprise vs equity value; FCFO vs FCFE	Exam, JJ Jones, all valuation methods
CORE	Reorganizing financial statements	Turn accounting into valuation inputs
PRO	Enterprise-to-Equity Value bridge	Interviews, project, real valuation work
TECH	Statement reconciliation and adjustments	Avoid model errors
PRACTICE	Discussion question	Convert formulas into skill

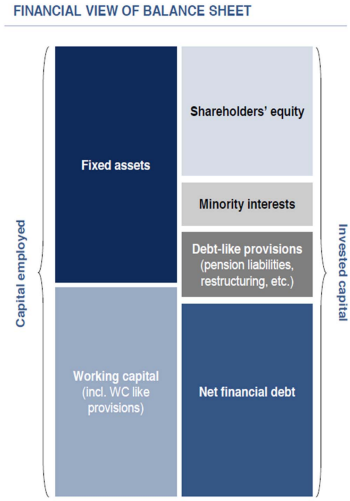
Valuation Procedures

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Starting Point: Profit & Loss and Balance Sheet





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Valuation Methods: One Map

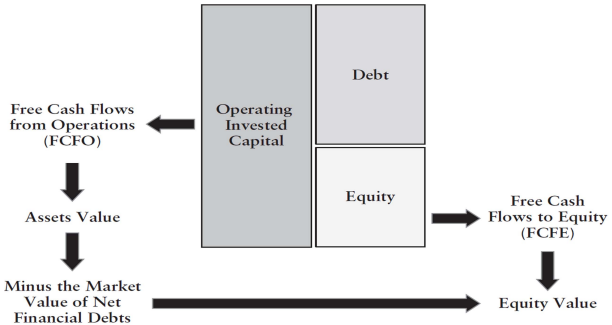
Approach	Main Question	Examples
Income Approach	What are future cash flows worth today?	DCF, DDM, FCFO, FCFE
Market Approach	What do comparable firms/transactions imply?	Trading multiples, deal multiples
Asset Approach	What are assets minus liabilities worth?	Net asset value, liquidation value
Economic Profit	Does the firm earn above its cost of capital?	EVA, residual income

- In this course, we focus on **income** and **market** approaches ...

  ... but all methods require clean financial statements

Valuation Results: Enterprise Value to Equity Value Bridge

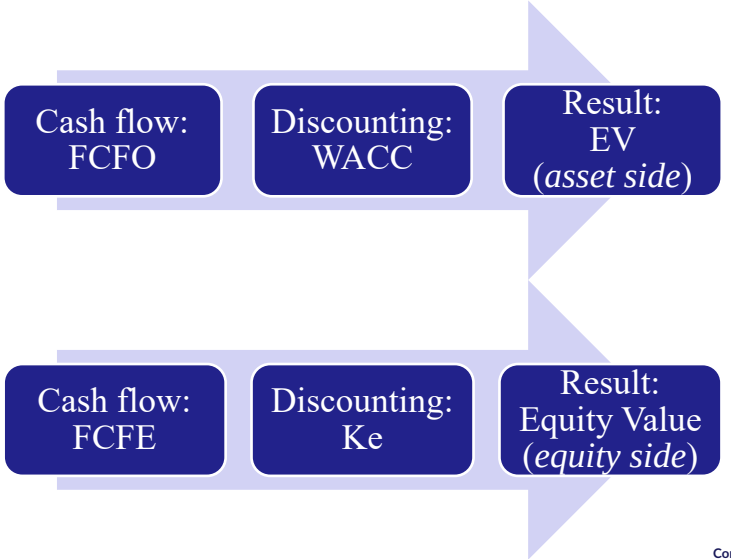
CORE



Equity Value + Net Debt = Enterprise Value (Assets Value)

Valuation Perspective: Asset Side vs Equity Side

CORE



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The Matching Principle of Valuation

Cash Flow	Includes Net Debt Issues/Repayment?	Discount Rate	Value
FCFO/FCFF	No	WACC	Enterprise Value
FCFE	Yes	Cost of Equity	Equity Value

Bottom line: Never mix FCFO with cost of equity; Never mix FCFE with WACC

Valuation Results: Enterprise Value to Equity Value Bridge

PRO

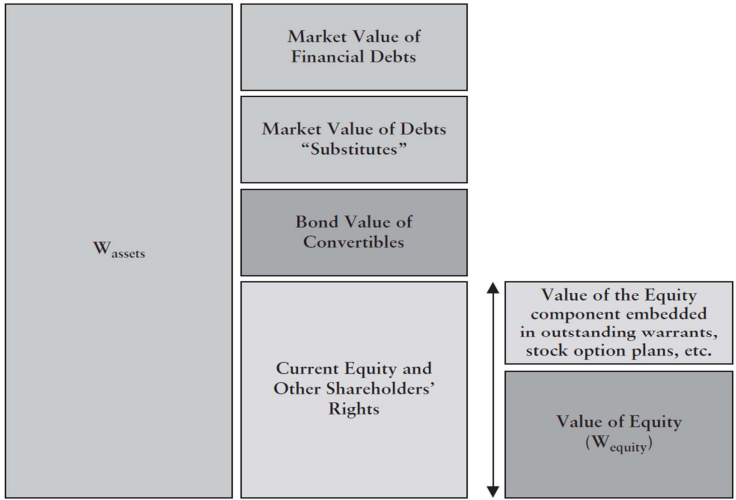
ENTERPRISE VALUE	
• **Net Financial Debt**	• **Net Financial Debt** = **Financial Debt** − **Cash & Cash Equivalent** • It is common practice to use book value of debt when computing the net debt of a company. • However, think of market value of debt when applicable, if all or part of debt is listed or traded over the counter or by discounting future debt flows at market cost of debt
• **Pension Provisions**	• **Pension Provisions** = (**Projected Benefit Obligation** − **Fair Value of plan assets**) × (**1** − **Tax rate**) • For Defined Benefit regimes only • Take Pension provisions net of deferred tax in the balance sheet if PBO and FVA not available
• **Other Debt-Like Items**	• **Other debt-like items** = **Debt-like provisions** × (**1** − **Tax rate**) • Debt-like provisions (added to net debt) may include restructuring provisions (if not recurrent), environmental damage, certain claims and litigations • See detailed footnotes relating to provisions and see which ones are tax deductible
• **Non-Operating Assets**	• Include financial stakes in associates, financial stakes in JVs (if not related to the Company's business and not in cash flows), other non-operating assets (land, buildings) • Use fair value of assets if possible (if listed companies, if market values known, ...), book value, P/E, or P/B ratio
• **Minority Interests**	• Minority interests may be valued at fair value At least by applying a PB ratio to the equity value of the company, or at book value
EQUITY VALUE	

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Valuation Results: Enterprise Value to Equity Value Bridge

PRO



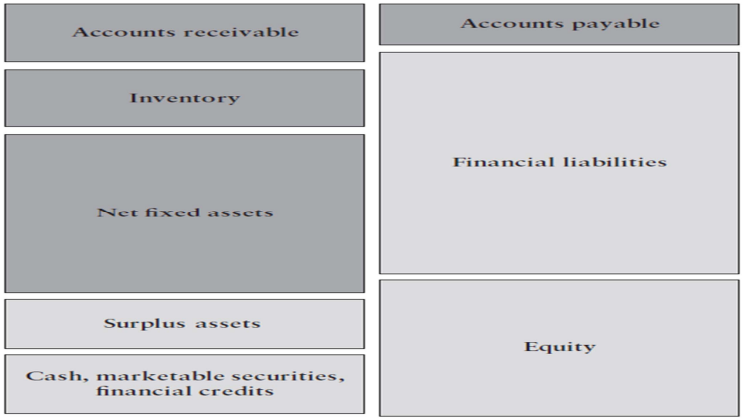
Earnings and Cash Flows

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Reorganization of Financial Reports: Balance Sheet

- To construct cash flows, we use the *functional* reorganization of the balance sheet:
  Aggregation of assets and liabilities based on the business operations they belong to



Accounting classifies items for reporting; valuation reorganizes them by economic function

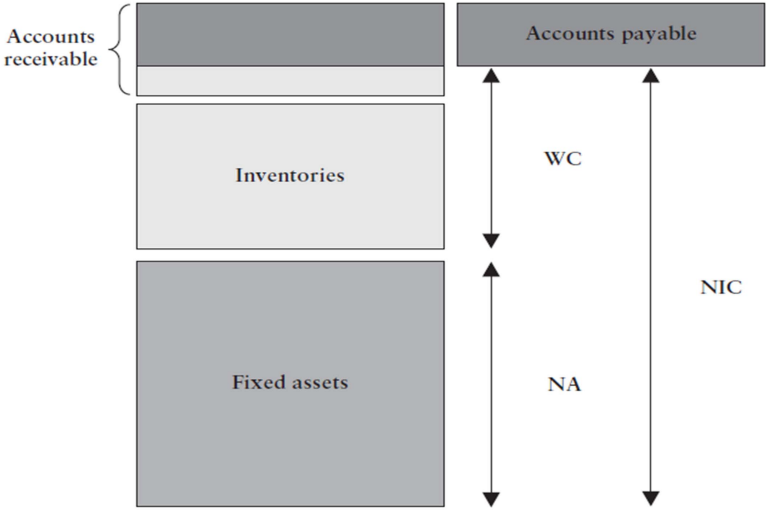
Reorganization of Financial Reports: Balance Sheet

- The categories related to the cycle of business operations include:
  - + Inventory
  - + Accounts Receivable
  - − Accounts Payable (for goods/services related to operations)

The algebraic sum of these categories yields the operating Net Working Capital (NWC)

- The net (operating) invested capital (NIC) is composed of:
  1. Net (Operating) Assets (i.e., operating assets net of D&A expenses and impairment losses)
     - The size of the fixed assets, relative to business volume, depends on the capital intensity index and therefore on the business-technical characteristics of the productivity process
  2. Working Capital
     - The size of the working capital, keeping the business volume fixed, is directly related to the length of the business operations' financial cycle (buy-transform-sell)

Reorganization of Financial Reports: Balance Sheet



Reorganization of Financial Reports: Balance Sheet

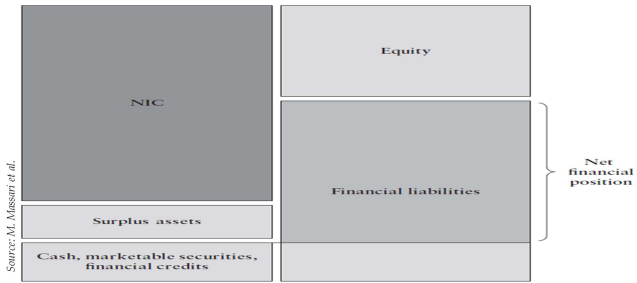
- One benefit of the functional reorganization model: precise link between the accounting values in the balance sheet and their respective flows in the income statement (see next slide). Establishing such a precise link allows us to
  - Make a rigorous analysis of profitability
  - Simplify the construction of economic and financial plans

- Establishing a specific category of revenues and costs for non-core activities allows us to separate the valuation of:
  (i) The company's core assets and
  (ii) Surplus assets

- To evaluate the company as a sum of parts (SOP), it is necessary to reconstruct the costs relative to each (separate) business area, normally up to the EBIT level

Reorganization of Financial Reports: Balance Sheet

- The net (operating and non-operating) assets are covered by:
  i. Financial liabilities negotiated by the company (in the short- and medium/long-term); and
  ii. Equity



- The proportion of net financial position (financial liabilities − cash & cash equivalent) w. r. to equity determines the financial structure of the company

Reorganization of Financial Reports: Income Statement

Why we care: FCFO starts from operating income, not net income

Income Statement at *Added Value*

1. **Sales Revenues, or Production value achieved during the fiscal year**
   - Consumption of raw materials and components acquired from third parties
   - Costs of services acquired from outside (e.g., supplies, ads, etc.)
   - Other external expenses (e.g., administration and control, organization, finance)
2. **Added value**
   - Labor costs
3. **EBITDA (Earnings Before Interest, Taxes, Depreciation, and Amortization)**
   - Depreciation & Amortization and Provisions
4. **Operating Income or EBIT (Earnings Before Interest and Taxes)**
   - Financial Expenses (+ Financial revenues) / + Net result from non-core activities
5. **Gross Operating Income from Continuing Operations**
   - Extraordinary losses (+ Extraordinary gains)
6. **EBT (Earnings Before Taxes)**
   - Tax expenses
7. **Net Income**

From Earnings to Cash Flows: Introduction

CORE

- Our goal is to consider all monetary inflows and outflows for all business areas
  - These flows, jointly, determine the firm's financial dynamics in a certain period
- The forces underlying a firm's financial dynamics include:
  - Continuing business operations and related investment/divestment in net working capital
  - Investment and divestment in fixed assets
  - Underwriting and repayment of financing instruments (including issuance of equity capital)
  - Payment of interests/dividends to debt/equity-holders
  - Net flows for the purchase or from the disposal of surplus assets
- The above forces can be reorganized as follows:
  1. Monetary flows of the operating area
  2. Monetary flows of the financing area
  3. Monetary flows of the non-core area

From Earnings to Cash Flows: Operating Area

CORE

- The cash generated from current business operations can be calculated by adding all costs that do not generate monetary transactions to the firm's operating result (mainly, D&A in industrial companies)

This figure is referred to as EBITDA

- EBITDA is not free cash flow: Working capital can absorb cash
- Note: can't always use EBITDA to finance outflows unrelated to current activities
  - The actual availability of cash flow generated by current activities in unrelated areas depends on the changes in NWC in the period under consideration
  - For instance, a positive change in NWC mirrors an employment of cash resulting in a partial absorption of EBITDA. Therefore:

$$\text{Monetary flow of current activities} = EBITDA \pm \Delta WC$$

From Earnings to Cash Flows: Financing Area

PRO

- The monetary flow coming from the financing area can be calculated on the basis of all monetary movements referable to debt and equity capital, as follows:
  - $-$ Passive interest payment
  - $-$ Dividend distribution
  - $+$ New debt negotiations
  - $-$ Expiring debt repayment
  - $+$ Equity increases
  - $-$ Equity repayment
  - $=$ Monetary flow from the financing area
- Note all these flows should be calculated net of fiscal adjustments for the company (not for debt- or equity-holders)

From Earnings to Cash Flows: Non-Core Area

PRO

- The monetary flow from surplus (non-core) assets is easily calculated due to the reduced number of elements that belong to this (residual) category
- In fact, this flow is mainly composed of two fundamental components:
  1. The income from non-core businesses
  2. the difference between:
     - New investments in non-core businesses; and
     - Divestments in non-core businesses
- Note all these flows should be calculated net of fiscal adjustments for the company (not for debt- or equity-holders)

TECH Layer: Reconciling the Statements

- The formulas are simple enough
- The model is harder

  Every item must reconcile across income statement, balance sheet, and cash-flow statement

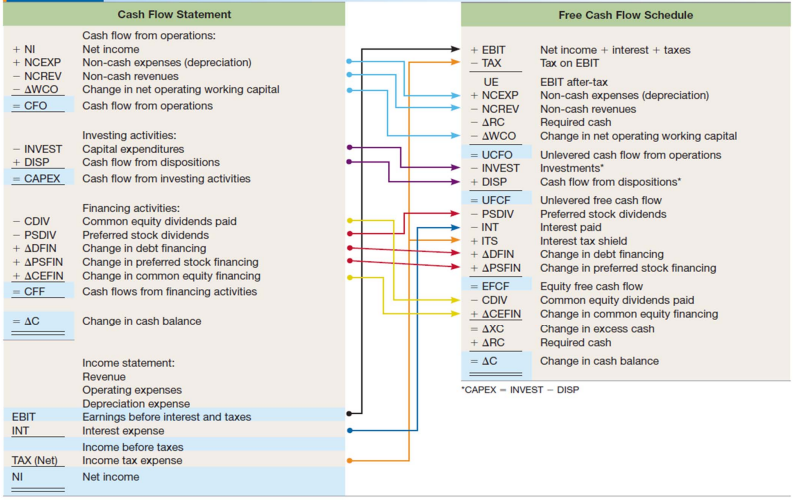
From Earnings to Cash Flows: Mechanics — Part 1

EXHIBIT 3.4 Relationships Between Changes in the Balance Sheet Accounts and the Cash Flow Statement

Change in Balance Sheet Accounts				Cash Flow Statement	
	Year 1	Year 2	Change	Year 2	
Balance Sheet					**Cash Flow from Operations**
Cash	C_{Y1}	C_{Y2}		+ NI	Net income
Current operating assets	COA_{Y1}	COA_{Y2}		+ NCEXP	Non-cash expenses (depreciation)
Total current assets	TCA_{Y1}	TCA_{Y2}		− NCREV	Non-cash revenues
Net property, plant, and equip.	$PPEQ_{Y1}$	$PPEQ_{Y2}$		− ΔWCO	Change in net operating working capital
Non-current operating assets	$NCOA_{Y1}$	$NCOA_{Y2}$		= CFO	Cash flow from operations
Total assets	TA_{Y1}	TA_{Y2}			
					Investing Activities
Current operating liabilities	COL_{Y1}	COL_{Y2}		− INVEST	Capital expenditures
Interest payable	INT/P_{Y1}	INT/P_{Y2}		+ DISP	Cash flow from dispositions
Current portion LT debt	$CLTD_{Y1}$	$CLTD_{Y2}$		= CAPEX	Cash flow from investing activities
Total current liabilities	TCL_{Y1}	TCL_{Y2}			
Long-term debt	LTD_{Y1}	LTD_{Y2}			**Financing Activities**
Non-current operating liabilities	$NCOL_{Y1}$	$NCOL_{Y2}$		− CDIV	Common equity dividends paid
Total liabilities	TL_{Y1}	TL_{Y2}		− PSDIV	Preferred stock dividends
Preferred stock	PS_{Y1}	PS_{Y2}		+ ΔDFIN	Change in debt financing
Common stock	CS_{Y1}	CS_{Y2}		+ ΔPSFIN	Change in preferred stock financing
Retained earnings	RE_{Y1}	RE_{Y2}		+ ΔCEFIN	Change in common equity financing
Total shareholders' equity	SE_{Y1}	SE_{Y2}		= CFF	Cash flows from financing activities
Total liabilities and equities	$TLSE_{Y1}$	$TLSE_{Y2}$		= ΔC	Change in cash balance
		Year 2			
Statement of Retained Earnings					
Beginning-of-year balance		RE_{Y1}			
Net income		NI			
Preferred stock dividends		−PSDIV			
Common equity dividends		−CDIV			
End-of-year balance		RE_{Y2}			

From Earnings to Cash Flows: Mechanics — Part 2

EXHIBIT 3.8 The Relations between the Cash Flow Statement and Free Cash Flow Schedule



Two Cash Flow Definitions to Remember

FCFO = EBIT $\times$ (1 $-$ tax rate) + non-cash charges $-$ CapEx $-$ increase in non-cash WC

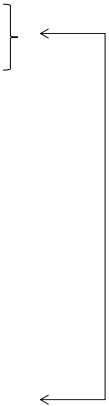
FCFE = net income + non-cash charges $-$ CapEx $-$ increase in non-cash WC + net debt issued

From Earnings to Cash Flows: Cash Flows Definitions

1. Free Cash flows to Equity (FCFE) — *Equity Side Valuation*
   - \+ Net Income — Adjusted
   - +/− Non Cash Expenditure (other than D&A) / Non Cash Revenues
   - − Net Capex (= Capital Expenditures − Depreciation/Amortization)
   - − Changes in non-cash Working Capital
   - − (Principal Debt Repayments − New Debt Issues)
   - = **Free Cash Flows to Equity**

2. Free Cash Flows from Operations (FCFO) — *Asset Side Valuation*
   - \+ FCFE
   - \+ Interest Expense (1 − Tax Rate)
   - \+ (Principal Debt Repayments − New Debt Issues)
   - = **Free Cash Flows from Operations (FCFO)**

   - \+ Operating Income (EBIT) — Adjusted
   - +/− Non Cash Expenditure (other than D&A) / Non Cash Revenues
   - − Taxes on EBIT
   - − Reinvestment Needs (Net Capital Expenditures and Working Capital)
   - = **Free Cash Flows from Operations (FCFO)**



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[Ignoring preferred dividends. If preferred stock exist, preferred dividends will also need to be netted out]

Discussion Questions

Consider a company with the following data

EBIT	$400
Interest Expensed and Paid	$100
Income Tax Rate on All Income	40%
Net Income	$180
Depreciation	$200
Capital Expenditures	$540
Increase in Required Non-Cash Working Capital	$80
Increase in Required Cash	$20
Increase in Debt Financing	$400
Paid-in Common Dividends	$90
Change in Cash Balance on Balance Sheet	$70
Cash Flow from Operations	$300

i. Prepare the Cash Flow Statement forecast

ii. Prepare the FCFO forecast

iii. Prepare the FCFE forecast

What These Discussion Questions Test

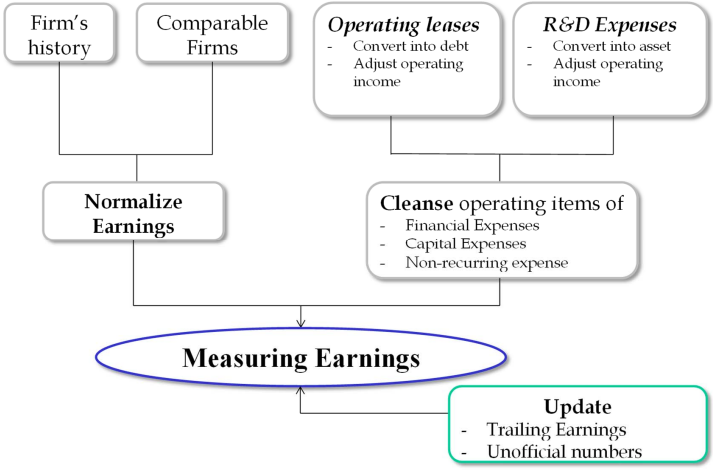
1. CFO is not FCFO
2. FCFO excludes (debt) financing
3. FCFE includes net debt issuance
4. Dividends are a use of FCFE, not part of FCFO
5. Required cash is not excess cash

From Earnings to Cash Flows: Cash Flows Definitions

- The concepts introduced so far apply to projections used in valuation techniques
- The outline presented in the previous slide highlights:
  1. The cash flows generated by the operating business (i.e., the **Free Cash Flow from Operations**, FCFO; also known as the **Free Cash Flow to Firm**, FCFF)
     - In the calculation of FCFO, the term taxes measures the theoretical fiscal outflow calculated on the operating result in a scenario of no debt
  2. The (net) cash flows available to shareholders (i.e., the free cash flow to equity, FCFE)
     - The value of the FCFE depends not only on the incidence of interest and income taxes but also on the change in the net financial position
     - Therefore, the projections set forth by the expert for the company's valuation need to include a target for the firm's financial structure(1)

(1): FCFE depends on the future financing operation choices involving the use of debt

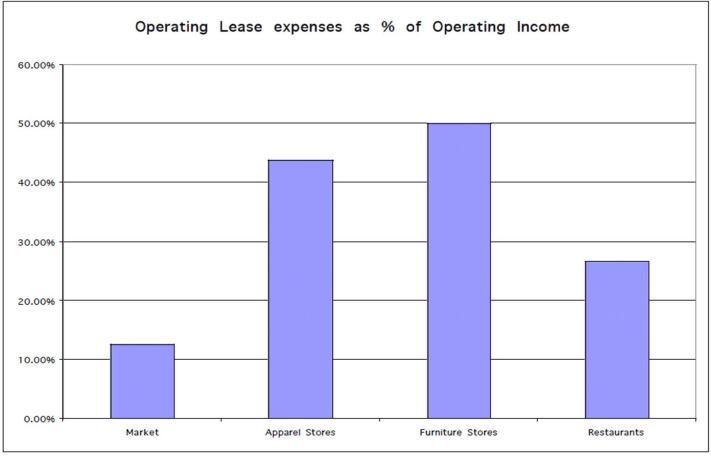
From Reported to Actual Earnings



From Reported to Actual Earnings: Adjusting Accounting Earnings

- Make sure there are no financial expenses mixed in with operating expenses
  - Financial expense: any tax deductible commitment that you have to meet no matter what your operating results. Failure to meet it leads to loss of control of the business
  - Example — Operating Leases: while accounting convention may treat operating leases as operating expenses, they are really financial expenses and need to be reclassified as such; this has no effect on equity earnings but does change the operating earnings
- Make sure that there are no capital expenses (CAPEX) mixed in with the operating expenses (OPEX)
  - Capital expense: any expense that is expected to generate benefits over multiple periods
  - Example — R & D Adjustment: Since R&D is a capital expenditure (rather than an operating expense), the operating income has to be adjusted to reflect its treatment

From Reported to Actual Earnings: Operating Leases



From Reported to Actual Earnings: Operating Leases

- Operating lease expenses may be treated as operating expenses in computing operating income
- More appropriately, operating lease expenses should be treated as financing expenses, with the following adjustments to earnings and capital:
  - Debt Value of Operating Leases: Present Value of operating lease commitments at pre-tax cost of debt
  - When you convert operating leases into debt, you also create an asset of the same value to counter it

Adjusted Operating Earnings

$$\underline{\text{Adjusted Operating Earnings}} = \text{Operating Earnings} + \text{Operating Lease Expense} - \text{Depreciation on Leased Asset}$$

As an approximation, this could work:

$$\underline{\text{Adjusted Operating Earnings}} = \text{Operating Earnings} + \text{Pre-tax cost of Debt} \times \text{PV of Operating Leases}$$

From Reported to Actual Earnings: Operating Leases

- GAP in 20XX has EBIT of $ 1,012, debt of $ 1.97 billion. Its pre-tax cost of debt is 6%
- Its operating lease payments were $ 978 million and its commitments were:

Year	Commitment ($ millions)	Present Value at 6%
1	899.00	$ 848.11
2	846.00	$ 752.94
3	738.00	$ 619.64
4	598.00	$ 473.67
5	477.00	$ 356.44
6 & 7	982.50 each year	$ 1,356.04
	Debt Values of leases =	$ 4,396.85 (Also value of leased asset)

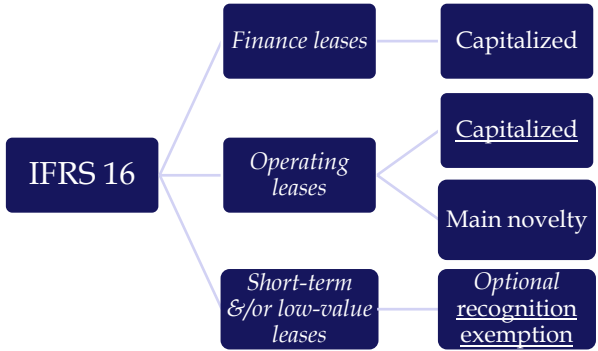
- Adjusted Debt Outstanding = $ 1,970m + $ 4,397m = $ 6,367m
- Adjusted EBIT = Reported EBIT + OL Exp. this year − Depreciations = $= \$1{,}012 + 978 - (4{,}397/7) = \$1{,}362$m (7 year useful life for assets)
- Approximate EBIT = $ 1,012 + $ 4,397 × 6% = $ 1,276m

From Reported to Actual Earnings: Operating Leases

Conventional Accounting	*Operating Leases Treated as Debt*
Income Statement EBIT before Leases = 1,990 Op Leases = 978 EBIT = 1,012 = (1,990 – 978)	***Income Statement*** EBIT before Leases = 1,990 - Depreciation Leased Assets = 628 Adjusted EBIT = 1,362 (1,990 – 628) *Interest expense will rise to reflect the conversion of operating leases as debt.*
Balance Sheet Off balance sheet (Not shown as debt or as an asset). Only the conventional debt of $1,970 million shows up on balance sheet	***Balance Sheet*** Asset: OL Asset 4,397 Liability: OL Debt 4,397 Total debt = 4,397 + 1,970 = $6,367 million
Cost of capital = 8.20% (7,350/9,320) + 4% (1,970/9,320) = 7.31% *Cost of equity for The Gap = 8.20%* *After-tax cost of debt = 4%* *Market value of equity = 7350*	**Cost of capital** = 8.20% (7,350/13,717) + 4% (6,367/13,717) = 6.25% *Cost of equity for The Gap = 8.20%* *After-tax cost of debt = 4%* *Market value of equity = 7350*
Return on Invested Capital = = 1,012(1-.35)/(3,130+1,970) = 12.90% *Book Value of Equity = 3,130* *Tax rate = 35%*	**Return on Invested Capital** = 1,362(1-.35)/(3,130+6,367) = 9.30% *Book Value of Equity = 3,130*

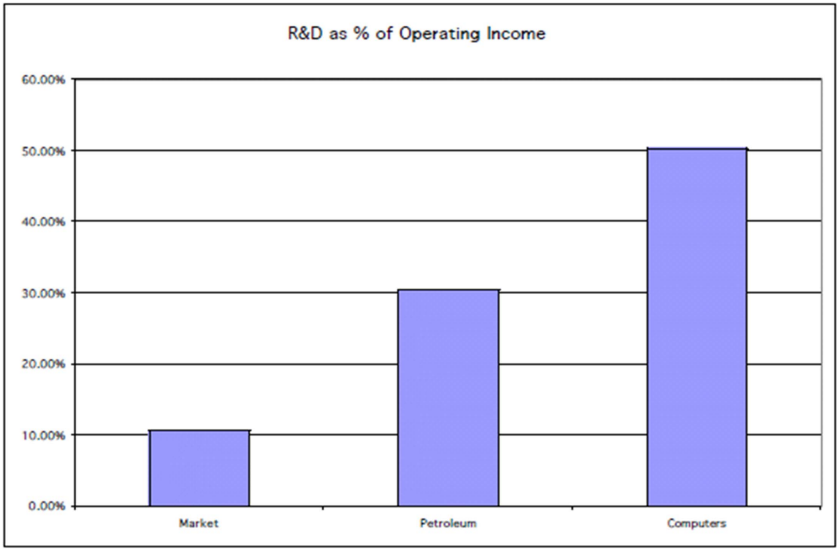
From Reported to Actual Earnings: Operating Leases

- In January 2016, IFRS 16 was issued superseding IAS 17
- IFRS 16 applies to reporting periods beginning on or after January 1st, 2019
- Under IFRS 16 all leases have to be capitalized (shown on the BS of the lessee)[1]



1: IFRS 16 provides exemption from capitalization for: (i) short-term leases and (ii) leases of low-value assets

From Reported to Actual Earnings: R&D Expenses



From Reported to Actual Earnings: R&D Expenses

- SAP 20XX Example (EBIT before R&D = 3,045; R&D assumed to have a 5-year life)

Year	R&D Expense	Unamortized portion		Amortization this year
Current	1020.02	1.00	1020.02	
−1	993.99	0.80	795.19	€198.80
−2	909.39	0.60	545.63	€181.88
−3	898.25	0.40	359.30	€179.65
−4	969.38	0.20	193.88	€193.88
−5	744.67	0.00	0.00	€148.93

- Value of research asset = €2,914m
  (assumption: sum of unamortized portions)
- Amortization of research asset in current year = €903m
  (sum of amortization this year per each asset's "layer")
- Increase in Operating Income = 1,020 − 903 = €117m

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From Reported to Actual Earnings: R&D Expenses

Conventional Accounting	*Capitalized R&D Expenses*
Income Statement EBIT before R&D = 3,045 - R&D expenses = 1,020 Reported EBIT = 2,025 Reported EBIT * (1-t) = 1,285 *Tax rate = 36.54%*	***Income Statement*** EBIT & R&D = 3,045 - Amortization R&D = 903 Adjusted EBIT = 2,142 (Increase of 117 m) Adjusted EBIT * (1-t) = 1,359 Ignored tax benefit = (1,020-903)*(0.3654) = 43 Adjusted EBIT after Tax Ben. = 1,359 + 43 = 1,402 (Increase of 117 million) Net Income will also increase by 117 million
Balance Sheet *Off balance sheet. Book value of equity at 3,768 million Euros is understated because one of the biggest assets is off the books.*	***Balance Sheet*** Asset Equity R&D Asset 2,914 Book Equity + 2,914 Adjusted Book Equity = 3,768 + 2,914 = 6,682
Capital Expenditures *Conventional net CAPEX of 2 million Euros*	***Capital Expenditures*** Adjusted Net CAPEX = 2 + 1,020 - 903 = 119
Cash Flows EBIT (1-t) = 1,285 - Net CAPEX = 2 FCFF = 1,283	***Cash Flows*** Adjusted EBIT * (1-t) = 1,402 - Adjusted Net CAPEX = 119 FCFF = 1,283
Return on Invested Capital = 1,285/(3,768+530) = 29.90% *Assumption: Liabilities = 530*	**Return on Invested Capital** = 1,402/(6,682+530) = 19.44% *Assumption: Liabilities = 530*

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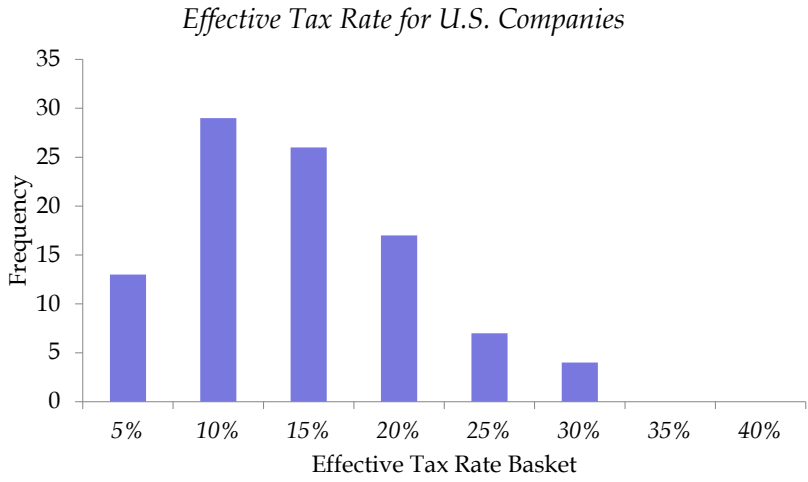
From Reported to Actual Earnings: R&D Expenses

- R&D expenses are tax deductible $\Rightarrow$ when capitalizing them, a tax benefit is missing
- Thus, we add the tax savings on the difference between the entire R&D Expense and the amortized amount of the research asset to the after-tax adjusted EBIT

$$
\begin{aligned}
\text{Adjusted EBIT After Tax}_t &= (\text{EBIT}_t + \text{R\&D Expenses}_t - \text{D\&A R\&D Asset}_t) \times (1 - \tau_C) \\
\text{Additional Tax Benefit}_t &= (\text{R\&D Expenses}_t - \text{D\&A R\&D Asset}_t) \times \tau_C \\
\text{Corrected Adjusted EBIT After Tax}_t &= \text{Adjusted EBIT After Tax}_t + \text{Additional Tax Benefit}_t \\
&= (\text{EBIT}_t + \text{R\&D Expenses}_t - \text{D\&A R\&D Asset}_t) \times (1 - \tau_C) + \\
&\quad + (\text{R\&D Expenses}_t \text{D\&A R\&D Asset}_t) \times \tau_C = \\
&= \text{EBIT}_t \times (1 - \tau_C) + \text{R\&D Expenses}_t - \text{D\&A R\&D Asset}_t
\end{aligned}
$$

When adjusting EBIT for R&D Expenses capitalization, we can use pre-tax R&D Expenses and pre-tax Estimated Amortization of R&D Asset

Tax Rate: Effective Tax Rates of US Firms



Tax Rate: Impact on Valuations (2018 Reform)

Macro Inputs
US T.Bond rate (1/1/18) = 2.41%
ERP = 5.08%

Tax Reform and Aggregate Value Effects: US Equities

	Pre-tax reform	*Post-tax reform*
Beta	1.07	1.07
Pre-tax cost of debt	3.91%	3.91%
Marginal Tax Rate	38.00%	24.00%
Debt to Capital Ratio	23.51%	23.51%
Revenues	$12,254.10	$12,254.10
Operating Income (EBIT)	$1,438.22	$1,438.22
Effective tax rate	25.19%	20.00%
After-tax return on capital	12.76%	13.65%
Reinvestment Rate =	59.27%	65.00%
Length of growth period =	5	5
Computed Values	Pre-tax reform	Post-tax reform
Cost of Equity =	7.85%	7.85%
After-tax cost of debt =	2.42%	2.97%
Cost of capital =	6.57%	6.70%
After-tax return on capital =	12.76%	13.65%
Reinvestment Rate =	59.27%	65.00%
Expected growth rate=	7.56%	8.87%
Value of firm		
PV of FCFF in high growth =	$2,253.08	$2,139.72
Terminal value =	$30,926.29	$34,590.66
Value of firm today =	**$24,750.46**	**$27,151.37**

Marginal Tax Rate
The drop in the federal corporate tax rate from 35% to 21% lowers overall marginal tas rate (with state & local taxes) from 38% to 24%

Effective tax rate
Change in corporate tax rate on US income & shift to regional tax model for global taxes will lower effective tax rate from 25.19% to 20%

Tax effect on debt
Lower marginal tax rate increases after tax cost of debt and capital (holding debt ratio fixed).

Higher ROIC/Reinvestment
ROIC rises proportionatelhy with drop in effective tax rate. Capital expensing rules lead to marginally more reinvestment.
Expected Growth = ROIC * Reinvestment Rate

Value with old tax code inputs = $24,751 billion
Value with new tax code inputs = $27,151 billion
Change in value = **$2,400 billion**
Percentage Change in value = 2400/24,751 = **9.70%**

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Tax Rate: Which One to Use (Summary)

- The choice is between effective and marginal tax rate

- **Doing projections, it is safer to use marginal tax rate** since the effective one simply reflects difference between accounting and the tax books

  1. Marginal tax rate. By using the marginal one, we tend to bias the after-tax operating income (NOPAT) in the earlier years, but we get it more accurate in later years
  2. Effective tax rate. If instead you choose the effective one (= Taxes Due / Taxable Income), then adjust it towards the marginal one over time, because tax-based revenues/expenses should converge to accruals (and thus cash) in the long-run (i.e. in perpetuity)

- When dealing with multinational companies, be careful. Either:
  - Use a weighted average of the marginal tax rates, with the weights based on the income derived by the firm from each country where it operates; or
  - Use the marginal tax rate of the country in which the company is incorporated, assuming implicitly the income will be eventually repatriated (**safest** approach); or
  - Keep the income from each country separate and apply a different marginal tax rate to each income statement

Reinvestment Needs

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Reinvestment Needs: Net Capital Expenditures I

- FCFE and FCFF are computed after reinvestments, which include Net CAPEX (long-term) and Non-Cash Working Capital (short-term)
  - Net CAPEX = CAPEX − D&A
  - Non-Cash Working Capital: removes Excess Cash (beyond day-to-day operations needs)

- As we estimate FCFE/FCFF starting from earnings after D&A (net income and EBIT), we need to add D&A back to obtain a cash flow measure:

$$\text{Cash Flows from Operations} = \text{EBIT} \times (1 - \tau_C) + \text{Non Cash Expenses (e.g., D\&A)} - \text{Non Cash Revenues}$$

- D&A have to be adjusted consistently with the adjustments made to earnings (D&A of capitalized operating leases, R&D expenses, ...)
- Net CAPEX is a function of how fast a firm is growing or expecting to grow (high growth firms have much higher net CAPEX than low growth ones)
- Assumptions about net CAPEX can therefore never be made independently of assumptions about future growth

Reinvestment Needs: Net Capital Expenditures II

When estimating Net CAPEX, we need to account for:

1. Need for smoothing lumpy CAPEX: firms seldom have smooth CAPEX. We need to normalize them:

$$\text{Estimated Net CAPEX} = \frac{1}{n}\sum_{i-1}^{n} \text{CAPEX}_{t-i} - \text{D\&A}_t$$

Important caveats:
(i) Do not average D&A as they are already smoothed
(ii) Choose carefully the number of historical years to use when averaging CAPEX
(iii) Check industry average ratios of (a) CAPEX/D&A (preferred) and/or (b) CAPEX/Sales

2. Acquisitions of other firms. Since these are like CAPEX, we need to normalize them:

$$\text{Adjusted Net CAPEX} = \text{Net CAPEX} + \text{Acquisitions of Other Firms} - \text{D\&A of Such Acquisitions}_t$$

Important Caveats:
(i) Most firms do not do acquisitions every year. Hence, should normalize the measure of acquisitions (e.g., look at an average over time)
(ii) May ignore acquisitions if you think they have been done at fair value (i.e. zero NPV investment accounting also for synergies) but be careful when estimating growth (growth from acquisitions should be excluded)

Reinvestment Needs: Net Capital Expenditures III

When estimating Net CAPEX, we need to account for:

3. CAPEX accounted as OPEX: if we decide to re-categorize some OPEX as CAPEX, we should estimate consistently the Adjusted Net CAPEX:
   a) Adjust (Gross) CAPEX:

$$\text{Adjusted CAPEX}_t = \text{CAPEX}_t + \text{R\&D Expenses}_t$$

   b) Adjust D&A:

$$\text{Adjusted D\&A}_t = \text{D\&A}_t + \text{R\&D Asset Amortization}_t$$

   c) Adjust Net CAPEX:

$$\begin{aligned}\text{Adjusted Net CAPEX}_t &= \text{Adjusted CAPEX}_t - \text{Adjusted D\&A}_t \\ &= \text{CAPEX}_t + \text{R\&D Expenses}_t - \text{D\&A}_t - \text{R\&D Asset Amortization}_t \\ &= \text{Net CAPEX}_t + \text{R\&D Expenses}_t - \text{R\&D Asset Amortization}_t\end{aligned}$$

- The adjustment made to Net CAPEX mirrors the adjustments made to EBIT
- We need to make consistent adjustments to operating earnings and to Net CAPEX

Reinvestment Needs: Changes in Non-Cash Working Capital

- The second component of reinvestment is the cash for working capital needs
- Definition of Non-Cash Working Capital (WC):
  1. Remove cash and investment in marketable securities, in excess from cash you need to run day-to-day operations: such cash amount should be either (i) deducted from the debt value when estimating the cost of capital or (ii) included as a surplus asset in the valuation
  2. Remove interest-bearing debt which is included in current liabilities, because it has to be considered when computing debt value to estimate the cost of capital
- Changes in Non-Cash WC from year to year tend to be volatile
  - A better estimate of Non-Cash WC could be made by looking at Non-Cash WC as a % of revenues (for the firm and industry)
- Some firms have negative Non-Cash WC
  - If this continues into the future it will generate positive cash flows for the firm
  - While this is feasible for some time, it is not feasible forever. Thus, it is better to set Non-Cash WC to zero in perpetuity, when it is negative over the first years of estimates

Reinvestment Needs: Volatility of Changes in Working Capital

	Amazon	Cisco	Motorola
Revenues	\$1,640	\$12,154	\$30,931
Non-Cash WC	−\$419	−\$404	\$2,547
% of Revenues	−25.53%	−3.32%	8.23%
Change from last year	−\$309	−\$700	−\$829
Average: last 3 years	−15.16%	−3.16%	8.91%
Average: industry	8.71%	−2.71%	7.04%

Assumptions in Valuation			
WC as % of Revenue	3.00% (?)	−3.32%	8.23%
Δ WC as % of Revenue in perpetuity	25.53% (?)	0%	0%

Common Mistakes in Converting Earnings to Cash Flows

1. Treating EBITDA as free cash flow
2. Including interest expense in FCFO
3. Forgetting that increases in NWC absorb cash
4. Treating all cash as excess cash
5. Discounting FCFO with cost of equity or FCFE with WACC
6. Adjusting EBIT for leases or R&D but forgetting the matching balance-sheet / CapEx adjustment
7. Moving from EV to equity value using only net debt when there are other claims

Takeaways

- Enterprise value and equity value are different valuation objects

$$\text{Enterprise Value (Assets Value)} = \text{Equity Value} + \text{Net Debt}$$

- FCFO values the business; FCFE values the equity
- Cash flows $\neq$ Earnings
  - This requires adjusting for non-cash expenses and revenues, changes in WC, CAPEX, and changes in the company's financing, similar to those made to prepare a cash flow statement
- Financial statements must be reorganized: operating, financing, non-core components
- Most common errors: mixing cash-flow definitions, discount rates, value objects
- Income tax disclosures is key to assess the value of a company
  - Average income tax rate based on company's books $\neq$ income tax rate actually paid
  - Income tax expense $\neq$ income taxes paid, used to measure free cash flows
  - Differences stem from permanent and temporary factors, which reverse